

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 15: FIRST ORDER DIFFERENTIAL EQUATIONS

1. (b) Find the solution of the differential equation

$$\frac{dy}{d\theta} + y \cot \theta = \sin^3 \theta$$

for which $y = 0$ when $\theta = \frac{1}{2}\pi$. $\left(y \sin \theta = \frac{1}{8} \left(\frac{1}{4} \sin 4\theta - 2 \sin 2\theta + 3\theta - \frac{3}{2}\pi \right) \right)$ [6]

Oct/Nov 2020/21/23/6 (part)

2. Find the solution of the differential equation

$$x \frac{dy}{dx} + 2y = e^x$$

for which $y = 3$ when $x = 1$. Give your answer in the form $y = f(x)$. $\left(y = \frac{(x-1)e^x+3}{x^2}\right)$ [8]

Oct/Nov 2020/22/4

3. Find the solution of the differential equation

$$\frac{dy}{dx} + 5y = e^{-7x}$$

for which $y = 0$ when $x = 0$. Give your answer in the form $y = f(x)$. $\left(y = \frac{1}{2}e^{-5x} - \frac{1}{2}e^{-7x}\right)$ [6]

May/June 2020/21/22/1

4. (a) Show that an appropriate integrating factor for

$$(x^2 + 1) \frac{dy}{dx} + y\sqrt{x^2 + 1} = x^2 - x\sqrt{x^2 + 1}$$

is $x + \sqrt{x^2 + 1}$. [4]

- (b) Hence find the solution of the differential equation

$$(x^2 + 1) \frac{dy}{dx} + y\sqrt{x^2 + 1} = x^2 - x\sqrt{x^2 + 1}$$

for which $y = \ln 2$ when $x = 0$. Give your answer in the form $y = f(x)$. $\left(y = \right)$ [7]

$$\frac{\ln 2 - \frac{1}{2} \ln(x^2 + 1)}{x + \sqrt{x^2 + 1}} = \left(x - \sqrt{x^2 + 1}\right) \ln \left(\frac{1}{2} \sqrt{x^2 + 1}\right)$$

May/June 2020/23/7

5. Find the solution of the differential equation

$$x \frac{dy}{dx} + 3y = \frac{\sin x}{x}$$

for which $y = 0$ when $x = \frac{1}{2}\pi$. Give your answer in the form $y = f(x)$. ($y = \frac{-x \cos x + \sin x - 1}{x^3}$) [8]

Specimen Paper 2020/2/3

6. (a) Find y in terms of x , given that

$$(x^3 + 8)^{\frac{2}{3}} \frac{dy}{dx} + x^2 y = 3x^2 (x^3 + 8)^{-\frac{1}{3}} e^{-(x^3+8)^{\frac{1}{3}}}$$

and that $y = 1$ when $x = 0$. ($y = e^{-(x^3+\frac{1}{8})^{\frac{1}{3}}} \left(\ln \left| \frac{1}{8}x^3 + 1 \right| + e^2 \right)$) [6]

Dec 1999/1/5 (part)

7. (a) Given that

$$(2x + 1) \frac{dy}{dx} + \frac{2}{3}y = \frac{(2x + 1)^{\frac{2}{3}}}{3x + 1},$$

find the general solution for y in terms of x . ($y = \frac{1}{3}(2x + 1)^{-\frac{1}{3}} \ln(3x + 1) + c(2x + 1)^{-\frac{1}{3}}$) [5]

Dec 1998/1/5 (part)

8. (a) (i) Use the result

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

to find $\frac{d}{dx} (\tanh^{-1} x)$. ($\frac{1}{1-x^2}$) [2]

(ii) Find $\frac{d}{dx} [\ln(\tanh^{-1} x)]$. ($\frac{1}{(1-x^2) \tanh^{-1} x}$) [1]

(iii) Hence, or otherwise, find the general solution of

$$\frac{dy}{dx} + \left[\frac{2}{(1-x^2) \tanh^{-1} x} \right] y = \frac{3x^2}{(\tanh^{-1} x)^2}. \quad [4]$$

$$(y = (x^3 + c)(\tanh^{-1} x)^2)$$

Dec 1997/1/5 (part)

9. (b) Find the complementary function of the differential equation

$$\frac{dy}{dx} + 2y = x + 1. \quad [2]$$

Find also a particular integral for this differential equation, and hence write down the general solution.

[4]

Nov 1996/2/14 (part)

10. (a) Find y in terms of x , given that

$$x \frac{dy}{dx} + (1 + x \tanh x)y = 1$$

and that $y = 0$ when $x = \ln 2$. ($xy \cosh x = \sinh x - \frac{3}{4}$)

[6]

Dec 1996/1/5 (part)

11. Water starts pouring into an empty open tank whose capacity is 1000 litres, and t seconds later the volume, V litres, of water in the tank is such that

$$\frac{dV}{dt} + \frac{1}{20}V = \frac{1}{4}t + 1.$$

- (i) Find V in terms of t , and hence show that, for large values of t ,

$$V \approx 5T - 80.$$

[6]

- (ii) Hence determine, to the nearest second, the value of t when the tank begins to overflow.

[1]

Jun 1996/1/6

12. (a) Obtain the solution, for y in terms of x , of the differential equation

$$\frac{dy}{dx} - (4 \tanh x)y = \sinh 2x,$$

given that $y = 0$ when $x = 0$. ($y = \frac{1}{4} \sinh^2 2x$)

[6]

Nov 1995/1/5 (part)

13. (a) Obtain y in terms of x , given that

$$(\sin x) \frac{dy}{dx} + (\cos x - 2x \sin x)y = 3x^2 e^{x^2} \quad (0 < x < \pi),$$

and that $y = 0$ when $x = 1$. ($y \sin x = e^{x^2}(x^3 - 1)$)

[5]

Jun 1995/1/5(part)

14. (a) By considering

$$\frac{d}{dx} [\ln\{(x+1)^2(2x-1)^3\}],$$

or otherwise, solve the differential equation

$$(2x^2 + x - 1) \frac{dy}{dx} + (10x + 4)y = \frac{1}{2x^2 + x - 1},$$

[7]

given that $y = 1$ when $x = 1$. $((x + 1)^2(2x - 1)^3 y = x^2 - x + 4)$
Dec 1994/1/5 (part)

15. (a) Find the general solution of the differential equation

$$(\coth x + 2\operatorname{cosech} x) \frac{dy}{dx} + y = x$$

[5]

where $x \neq 0$. $(y(2 + \cosh x) = x \cosh x - \sinh x + c)$
June 1994/1/5 (part)

16. (b) Find the solution of the differential equation

$$\frac{dy}{dx} + \frac{x+1}{\sqrt{x^2+1}} y = \frac{e^{x-\sqrt{x^2+1}}}{x+\sqrt{x^2+1}}.$$

[7]

given that $y = 5$ when $x = 0$. $\left(y = \frac{e^x + 5e^{-1}}{(x+\sqrt{x^2+1})e^{\sqrt{x^2+1}}}\right)$
Dec 1993/1/5 (part)

17. (a) Obtain the solution of the differential equation

$$(\sin x) \frac{dy}{dx} - (\cos x)y = 1,$$

[5]

given that $y = 2$ when $x = \frac{1}{2}\pi$.
Jun 1993/1/5 (part)

18. (a) Show that

$$\frac{2+x+x^2}{(1+x)(1+x^2)} = \frac{1}{1+x} + \frac{1}{1+x^2}.$$

[1]

Obtain the solution of the differential equation

$$(1+x)(1+x^2) \frac{dy}{dx} + (2+x+x^2)y = 1,$$

[6]

given that $y = 2$ when $x = 0$. $\left(y = \frac{1}{1+x} \left(1 + e^{-\tan^{-1} x}\right)\right)$
Dec 1992/1/5 (part)

19. (a) In the differential equation

$$x \frac{dy}{dx} + (xP - \alpha)y = x^{\alpha+1}Q \quad (x > 0),$$

α is a real constant while P and Q depend on x . Show that the substitution $y = x^\alpha z$ reduces the given differential equation to [3]

$$\frac{dz}{dx} + Pz = Q.$$

Hence, or otherwise, find the general solution for y in terms of x in the case where $P = \frac{x^2}{1+x^3}$ and $Q = 4x^2$. $\left(y = x^\alpha \left[(1+x)^3 + \frac{c}{(1+x^3)^{\frac{1}{3}}} \right] \right)$ [4]

Jun 1992/1/5 (part)

20. (a) Verify that

$$\frac{d}{dx} [\ln(x + \ln x)] = \frac{1+x}{x(x + \ln x)}.$$

Hence, or otherwise, find the solution of the differential equation

$$(x^2 + x \ln x) \frac{dy}{dx} + (1+x)y = 3x^3,$$

given that $y = 2$ when $x = 1$.

[6]
Dec 1991/1/5 (part)

21. (a) Find the general solution of the differential equation

$$x(1+x^2) \frac{dy}{dx} + (1+2x^2)y = 2x\sqrt{1+x^2} \quad (x > 0). \quad [6]$$

$$\left(y = \frac{x}{\sqrt{1+x^2}} + \frac{c}{x\sqrt{1+x^2}} \right)$$

Jun 1991/1/5 (part)

22. (a) Show, by means of the substitution $y = \frac{1}{z}$, that the differential equation

$$(1+x) \frac{dy}{dx} - 2y + (1+x)y^2 = 0$$

reduces to

$$\frac{dz}{dx} + \left(\frac{2}{1+x} \right) z = 1.$$

Hence find y in terms of x , given that $y = 1$ when $x = 0$.

$$\left(y = \frac{3(1+x)^2}{(1+x)^3 + 2} \right)$$

[7]

Jun 1990/1/5 (part)

23. (a) Solve the differential equation

$$\frac{dy}{dx} - (\cot x)y = \sin x \quad (0 < x < \pi),$$

given that $y = \pi$ when $x = \frac{1}{2}\pi$. $(y = (x + \frac{\pi}{2}) \sin x)$ [5]

Dec 1990/1/5

24. (a) Find the general solution of the differential equation

$$\frac{dy}{dx} + 3y = x,$$

giving y explicitly in terms of x in your answer. $(y = \frac{1}{3}x - \frac{1}{9} + Ce^{-3x})$ [5]

Nov 1988/2/14 (part)

25. (i) Obtain the solution of the differential equation

$$\frac{dy}{dx} + (\tanh x)y = 1$$

such that $y = 1$ when $x = 0$. $(y = \tanh x + \operatorname{sech} x)$ [6]

June 1988/1/5 (part)

26. (a) Find the solution of the differential equation

$$x \frac{dy}{dx} - y = x^3$$

such that $y = 1$ when $x = 1$. [6]

FM June 1988/1/5

27. (a) By considering $\frac{d}{dx}(x^2y \ln x)$, or otherwise, find y in terms of x given that $(x \ln x) \frac{dy}{dx} + (1 + 2 \ln x)y = 1$ ($x > 1$), and that $y = 1$ when $x = e$.

FM June 1987/1/5 (part)

28. (a) By using the substitution $z = y^2$, or otherwise, obtain the solution of the differential equation $(1 + x^2)y \frac{dy}{dx} + 2xy^2 = 3$ ($y \geq 0$) such that $y = 1$ when $x = 0$. Hence find the value of y when $x = 1$.

FM Nov 1987/1/5 (part)

29. (a) Given that $(\sin x) \frac{dy}{dx} - (\cos x)y = \sin^3 x$, and that $y = 1$ when $x = \frac{1}{2}\pi$, show that $y = \frac{1}{2}(1 + \sqrt{2})$ when $x = \frac{3}{4}\pi$.

FM June 1986/1/5

30. (a) Solve the differential equation $\frac{dy}{dx} + \left(\frac{e^x}{4+e^x}\right)y = 1$, given that $y = 1$ when $x = 0$.
 $\left(y = \frac{4+4x+e^x}{4+e^x}\right)$

FM Nov 1986/1/5 (part)

31. Find the general solution of the differential equation

$$x \frac{dy}{dx} + 4y - 10x = 0,$$

giving y explicitly in terms of x . Find also the particular solution such that $y \rightarrow 0$ as $x \rightarrow 0$. [6]
 $\left(y = 2x + \frac{c}{x^4}; y = 2x\right)$

Nov 1985/1/22 (part)

32. (a) Find the general solution of the differential equation

$$\frac{dy}{dx} = x + y,$$

giving y in terms of x in your answer. $(y = Ae^x - x - 1)$ [5]

Find also the particular solution for which the minimum value of y is zero. $(y = e^x - x - 1)$ [3]

June 1985/1/22 (part)

33. (b) Find y in terms of x given that

$$x \frac{dy}{dx} = 1 - 2y. \quad [5]$$

$$\left(y = \frac{1}{2} \left(1 - \frac{A}{x^2}\right)\right)$$

June 1985/2/7 (part)

30. (a) Obtain the general solution of the differential equation

$$\frac{dy}{dx} + \frac{4y}{1-4x^2} = 1 - 2x.$$

$$(y = (x^2 + x + c)(1 - 2x)(1 + 2x))$$

FM June 1985/1/5 (part)

34. (b) Find the general solution of the differential equation $(a^2 - x^2) \frac{dy}{dx} - xy + a^2 = 0$, where a is a positive constant and $|x| < a$. $\left(y = (Ax + B)e^{3x} + x + \frac{2}{3}\right)$

FM June 1985/2/5

35. Prove (do not merely verify) that the general solution of the differential equation

$$\frac{dy}{dx} + 2xy = 2x(1 + x^2)$$

may be expressed as

$$y = x^2 + Ae^{-x^2},$$

where A is an arbitrary constant.

[6]

June 1984/1/22 (part)

36. (b) Find y in terms of x given that $\frac{dy}{dx} = 3 - 2y$, and that $y = -1$ when $x = 0$. $\left(y = \frac{1}{2}(3 - 5e^{-2x})\right)$ [5]

June 1984/1/7 (part)

37. (a) Find the derivative of $\tan^{-1}\left(\frac{2}{x}\right) + \tan^{-1}\left(\frac{x}{2}\right)$, ($x > 0$). (0)
 (b) Find the general solution of the differential equation

$$(x^2 + 4)\frac{dy}{dx} = 2y + x^2 e^{\tan^{-1}\left(\frac{1}{2}x\right)},$$

expressing your answer in the form $y = f(x)$. Find also the particular solution for which $y = e^{\frac{1}{4}\pi}$ when $x = 2$. $\left(y = \left(x - \frac{1}{2}\tan^{-1}\left(\frac{1}{2}x\right) + c\right)e^{\tan^{-1}\left(\frac{1}{2}x\right)}; c = \frac{1}{2}\pi - 1\right)$

FM June 1984/1/4

38. (a) Find $\int e^{2x} \cos x \, dx$. $\left(\frac{1}{5}e^{2x}(2 \cos x + \sin x) + k\right)$
 (b) Solve the differential equation $\frac{dy}{dx} + 2y = \cos x$, given that $y = 1$ when $x = 0$. $\left(y = \frac{1}{5}(2 \cos x + \sin x + 3e^{-2x})\right)$

FM Nov 1984/2/6

39. (i) Find the general solution of the differential equation

$$x \frac{dy}{dx} - 2y + x = 0,$$

and find also the particular solution for which $y = 1$ when $x = 1$. $(y = x + Ax^2; y = x)$

June 1983/2/22 (part)

40. (b) Find the general solution of the differential equation $\frac{dy}{dx} + 2y \tan x = \sin x$. Given that $y = 0$ when $x = 0$, find the maximum value of y . $\left(y = \cos x + k \cos^2 x + c; y = \frac{1}{4}\right)$

FM June 1983/2/4

41. Express y in terms of x given that

$$(1 + x^2) \frac{dy}{dx} + xy = \sqrt{1 + x^2}$$

and $y = 1$ when $x = 0$. $\left(y = \frac{1+x}{(1+x^2)^{\frac{1}{2}}} \right)$

Prove that y has a maximum value when $x = 1$ and sketch the graph of y for $x \geq 0$.

Nov 1983/1/22

42. Find the general solution of the differential equation $\frac{dy}{dx} + 3y = \cos 2x$.

FM Nov 1983/2/5

43. (a) Find the general solution of the differential equation

$$t \frac{dx}{dt} - 3x = t^5,$$

giving x explicitly in terms of t . $\left(x = \frac{1}{2}t^5 + ct^3 \right)$

Nov 1982/1/22 (part)

44. (b) Find the general solution of the differential equation

$$(x^2 + 1) \frac{dy}{dx} + y = \tan^{-1} x.$$

$$\left(y = \tan^{-1} x - 1 + ke^{-\tan^{-1} x} \right)$$

FM Nov 1982/2/5 (part)

45. (a) Find the general solution of the differential equation

$$\frac{dy}{dx} - \frac{y}{x+1} = x,$$

giving an explicit expression for y in terms of x . $(y = (x+1)\{x - \ln(x+1) + c\})$

June 1982/2/22 (part)

46. (a) Given that $y = \tan(3 \tan^{-1} x)$, show that $(1 + x^2) \frac{dy}{dx} = 3(1 + y^2)$.

- (b) Solve the differential equation $x \frac{dy}{dx} = 2y + x^2 \ln x$, given that $y = 2$ when $x = 1$.
 $(y = x^3(\ln x - 1) + 3x^2)$

FM June 1982/1/5

47. (b) Find the general solution of the differential equation

$$\frac{dy}{dx} + y = 2x + 3.$$

$$(y = Ae^{-x} + 2x + 1)$$

Nov 1981/1/22 (part)

48. Obtain the general solution of the differential equation

$$x \frac{dy}{dx} - y = x^2 + 1$$

in the form $y = x^2 + Cx - 1$, where C is an arbitrary constant.

June 1981/2/22

49. (b) Solve the differential equation

$$x \frac{dy}{dx} + (x + 1)y = 2xe^{-x},$$

given that $y = 0$ when $x = 1$. $\left(y = \left(\frac{x-1}{x}\right)e^{-x}\right)$

FM Nov 1981/1/5 (part)

50. Find the general solution of the differential equation

$$\frac{dy}{dx} + 2xy = 2x(x^2 + 1).$$

FM June 1981/2/5 (part)

51. Find the general solution of the differential equation $\frac{dy}{dx} - 2y = 4x$, and find also the particular solution for which $y = -1$ when $x = 0$. $(y = ce^{2x} - 2x - 1; y = -2x - 1)$

June 1980/2/22

52. (a) Solve the differential equation

$$x \frac{dy}{dx} - 2y = x^4 \cos 2x.$$

$$\left(y = \frac{1}{2}x^3 \sin 2x + \frac{1}{4}x^2 \cos 2x + cx^2\right)$$

FM June 1980/1/4 (part)

53. Use the substitution $x = \frac{1}{y^2}$ to prove that the differential equation

$$x^2 \frac{dy}{dx} + xy = y^3$$

can be reduced to

$$\frac{dy}{dx} - \frac{2z}{x} = -\frac{2}{x^2}.$$

Hence, or otherwise, find the particular solution of the differential equation in y and x for

which $y = 1$ when $x = 1$. $\left(\frac{1}{y^2} = \frac{2}{3x} + \frac{x^2}{3}\right)$

Nov 1980/1/22

54. (b) By means of the substitution $z = \frac{1}{y^2}$, transform the differential equation $\frac{dy}{dx} + y = xy^3$ into a differential equation containing z and x . Solve this equation for z and hence show that the solution of the given differential equation for which $y = \sqrt{2}$ when $x = 0$ is $y = \left[\frac{2}{(2x+1)}\right]^{\frac{1}{2}} \cdot \left(\frac{dz}{dx} - 2z = 2x; z = Ae^{2x} + x + \frac{1}{2}\right)$ [7]

Specimen Paper Further Mathematics Syllabus C 9225/1/4

55. (a) Find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x+1}, (x > 0).$$

Find also the particular solution for which $y = 1$ when $x = 1$. $(xy = x \ln(1+x) + c; y = 1 + \frac{1}{x} \ln\left(\frac{2}{x+1}\right))$

June 1979/2/21

56. Prove that

(a) $\int \operatorname{sech} x \, dx = \tan^{-1}(e^x) + C,$

(b) $\int \operatorname{sech}^3 x \, dx = \frac{1}{2} \tanh x \operatorname{sech} x + \tan^{-1}(e^x) + C.$

where C is, in each case, an arbitrary constant.

Find the particular solution of the differential equation $\frac{dy}{dx} - 3y \tanh x = 1$ for which $y = 0$ when $x = 0$. $\left(y = \frac{1}{2} \sinh x \cosh x + \cosh^3 x \left(\tan^{-1} e^x - \frac{\pi}{4}\right)\right)$

Nov 1979/1/22

57. (a) Obtain the general solution of the differential equation

$$\frac{dy}{dx} = y \tan x + 2x \sec x.$$

$(y \cos x = x^2 + c)$
FM Nov 1979/1/7 (part)

58. Find y in terms of x if $\frac{dy}{dx} + \left(\frac{2}{x} - 2\right)y = -\frac{1}{x}, (x > 0)$. $\left(y = \frac{1}{2}x + \frac{1}{4}x^2 + \frac{ce^{2x}}{x^2}\right)$

Describe the behaviour of y as $x \rightarrow 0$ and as $x \rightarrow \infty$ in each of the following cases:

- (i) If $y = \frac{3}{4}$ when $x = 1$. ($y \rightarrow \infty$ as $x \rightarrow 0$ and $y \rightarrow 0$ as $x \rightarrow \infty$)
(ii) If $y = 1$ when $x = 1$. ($y \rightarrow \infty$ as $x \rightarrow 0$ and also as $x \rightarrow \infty$)

June 1978/1/22

59. (a) Find the general solution of the differential equation

$$t \frac{dx}{dt} + 3x = \sqrt{\left(3 + \frac{4}{t^2}\right)}.$$

$$\left(x = \frac{1}{9t^3} (3t^2 + 4)^{\frac{3}{2}} + \frac{c}{t^3}\right)$$

Nov 1978/1/22

60. Find the general solution of the differential equation $\frac{dy}{dx} + 2y = \sin 3x$.

June 1974/1/15 (part)

61. (a) Find the general solution of

$$x \frac{dy}{dx} + 2y = \frac{2 \sin x}{x \cos^3 x}.$$

$$(x^2 y = \sec^2 x + c)$$

FM June 1973/1/16 (part)

62. Find the general solution of the differential equation

(a) $x \frac{dy}{dx} - 3y = x^5 e^{-\frac{1}{2}x^2},$

FM Nov 1973/1/16 (part)